Online Marketplaces

Part 3: A/B Testing in Marketplaces

Bar Light

January 17, 2026

A/B Testing in Marketplaces

A/B testing has become a **popular method for experimentation** in online marketplaces to test features and strategies in a **data-driven way**. Its **simple** and relatively easy to implement.

This Lecture:

1. **A/B Testing: Introduction**
   - How A/B testing works. Class exercise.
2. **P-Values: Interpretation, misuse, and issues in marketplaces**
   - Continuous monitoring, multiple hypotheses.
3. **Network Interference**
   - Cluster Randomization and Switchback Experiments.
4. **Other Challenges in A/B Testing**
   - External validity, heterogeneity, experimenting on algorithms.

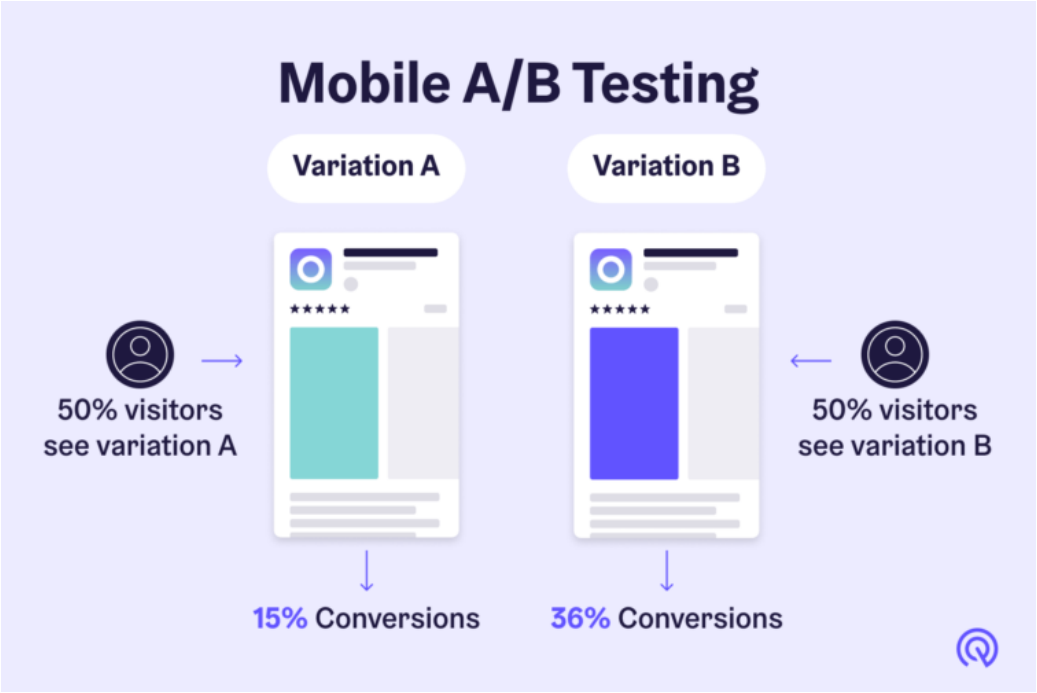
1. Introduction to A/B Testing

- A/B testing is a controlled experiment that compares two versions (A and B) of a product, feature, algorithm, etc to determine which performs better in achieving a specific goal.
- Widely used in online marketplaces, e.g., for:
  - Pricing experiments (e.g., Uber, Amazon).
  - UI/UX changes (Most Marketplaces).
  - Recommendation algorithms (e.g., Netflix).
- A/B testing helps in identifying "causal" relationships by splitting users into two groups:
  - **Control Group:** Receives the existing version (A).
  - **Treatment Group:** Receives the new version (B).

A/B testing are widely studied in statistics (e.g., sample size requirement, p-values, multiple testing, confidence interval).
Our focus: **marketplaces**.

A/B Testing Classic Example

- Variation A: Green "Buy Now" button.
- Variation B: Blue "Buy Now" button.
- Visitors are divided randomly between the two versions.
- **Purpose:** Identify which version achieves a better outcome (e.g., higher conversion rates).



Main Assumptions in A/B Testing

- **Random Assignment:** Users are randomly assigned to control or treatment groups.

  Randomization eliminates confounding bias.
- **Stable Unit Treatment Value Assumption (SUTVA):**
  - **No Interference Between Units:** One unit's treatment does not affect another unit's outcome (violated in marketplaces in many settings).

    No interference allows to treat each individual's outcome as being attributable solely to their own treatment.
  - **Consistency:** For each unit, there are no different forms or versions of each treatment level.

Main Assumptions in A/B Testing

- **No External Factors:** Factors outside the test design should not systematically impact the outcomes of one group differently from the other.
  Ensures observed differences can be attributed to the treatment.
- Other "statistical" properties (e.g., sufficient sample size).

Violation of Consistency

A/B Test Setup:
Testing two recommendation algorithms on an e-commerce platform to measure conversion rates:

- **Algorithm A:** Recommends popular items based on past sales.
- **Algorithm B:** Recommends personalized items based on user preferences.

Violation:

- Some users in Group A experience recommendations from Algorithm A mixed with Algorithm B due to incomplete rollout or caching issues.
- Others in Group A consistently experience recommendations from Algorithm A as intended.
- Treatment inconsistency within Group A violates SUTVA (consistency).

Violation of External Factors

A/B Test Setup:
Testing a new assortment algorithm to increase conversions.

- **Group A:** Users see assortments using the old algorithm.
- **Group B:** Users see assortments using a new algorithm.

Violation:

- A competitor launches a promotion halfway through the A/B test.
- The promotion attracts users disproportionately from Group A, reducing their conversions.

- External promotion introduces a bias in estimate against Group A.

Average Treatment Effect (ATE): Estimation

Definition:
The Average Treatment Effect (ATE) measures the **difference in outcomes** between treated and control groups, on **average**, across the population.

Estimation:

$$\widehat{\text{ATE}} = \frac{1}{N_t} \sum_{i \in \text{Treatment}} Y_i - \frac{1}{N_c} \sum_{j \in \text{Control}} Y_j$$

where:

- N_t: Number of individuals in the treatment group.
- N_c: Number of individuals in the control group.
- Y: Observed outcomes.

Average Treatment Effect (ATE): Example

Example Data:

Group	User ID	Outcome (Click: 1, No Click: 0)
Treatment	1	1
Treatment	2	0
Treatment	3	1
Treatment	4	1
Control	5	0
Control	6	0
Control	7	1
Control	8	0

Table: Observed Outcomes for Treatment and Control Groups

$$\widehat{\text{ATE}} = \frac{1}{4}(1+0+1+1) - \frac{1}{4}(0+0+1+0) = \frac{3}{4} - \frac{1}{4}$$

Interpretation: On average, the treatment increases the likelihood of a click by **50 percentage points**.

Standard Error of the Estimated ATE

Formula for Standard Error (unbiased estimator):

$$\mathrm{SE}(\widehat{\mathrm{ATE}}) = \sqrt{\frac{\widehat{\sigma_t}^2}{N_t} + \frac{\widehat{\sigma_c}^2}{N_c}}$$

- $\widehat{\sigma_t}^2 = \frac{1}{N_t - 1} \sum_{i \in \text{Treatment}} \left(Y_i - \overline{Y_t}\right)^2$: Sample variance in the treatment group where $\overline{Y_t}$ is the sample mean.
- $\widehat{\sigma_c}^2 = \frac{1}{N_c - 1} \sum_{i \in \text{Control}} \left(Y_i - \overline{Y_c}\right)^2$: Sample variance in the control group.
- N_t, N_c: Number of observations in the treatment and control groups.

Purpose:

The standard error quantifies the uncertainty in the estimate of the ATE.

Other measures of uncertainty?

Classic Statistical Testing: Hypotheses

Null Hypothesis (H_0):
The treatment has no effect: $\text{ATE} = 0$.

Alternative Hypothesis (H_1):
The treatment has an effect: $\text{ATE} \neq 0$.

(Normalized) Test Statistic:

$$z = \frac{\widehat{\text{ATE}}}{\text{SE}(\widehat{\text{ATE}})}$$

where:

- $\widehat{\text{ATE}}$: Observed difference in sample means.
- $\text{SE}(\widehat{\text{ATE}})$: Standard error of the estimate.

P-Values

P-Value:
The p-value is the probability of obtaining a test statistic at least as extreme as the one obtained from the data under the assumption that H_0 is true.

Computing p-values: Central Limit Theorem (CLT):

- Under **independence** assumptions, for sufficiently large sample sizes, z is approximately normal.
- Under the null hypothesis (H_0), the true ATE is 0.
- In this case, by CLT:

$$\frac{\widehat{\text{ATE}}}{\text{SE}(\widehat{\text{ATE}})} \sim \text{Normal}(0, 1) \quad \text{(approximately)}.$$

P-Value and Statistical Significance

P-Value Mathematical Representation:

$$p = P(|Z| > |z_{obs}|)$$

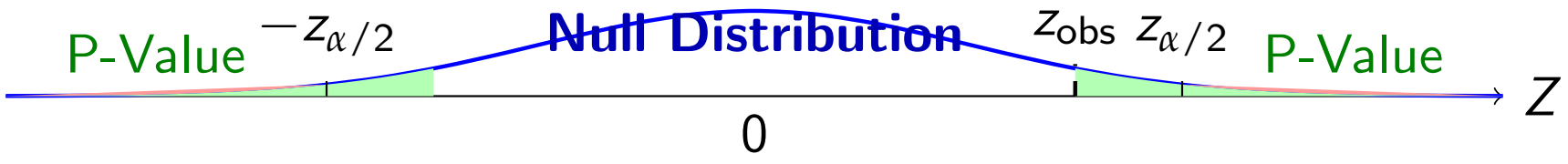
where Z follows a standard normal distribution and

$$z_{obs} = \frac{\widehat{\text{ATE}}_{\text{obs}}}{\text{SE}(\widehat{\text{ATE}}_{\text{obs}})}.$$

Decision Rule:

- Reject H_0 if $p < \alpha$ (e.g., 0.05): The result is statistically significant.
- Fail to reject H_0 if $p \geq \alpha$: No significant evidence of an effect.

P-Value and Statistical Significance

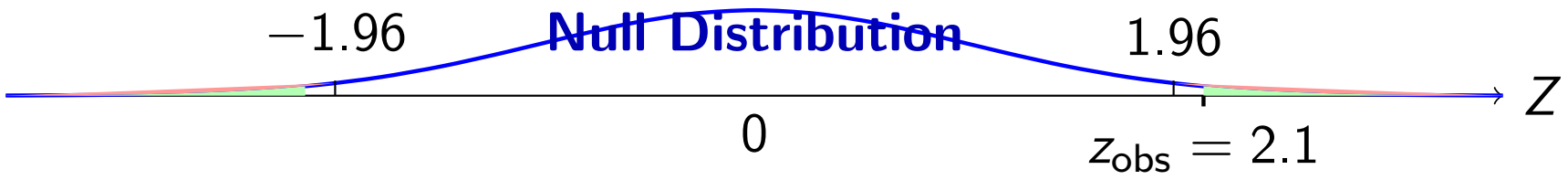


Explanation:

- Z: Standard normal random variable under the null hypothesis.
- $z_{\alpha/2}$: Critical values defining the rejection region for a significance level α.
- z_{obs}: Observed test statistic.
- **P-Value:** The green shaded areas correspond to the probability of observing a test statistic as extreme as z_{obs} under H_0.

Conclusion: In this case, z_{obs} is **not in the rejection region**, so we **fail to reject** H_0.

Numerical Example: Rejecting H_0



Numerical Explanation:

- A significance level of $\alpha = 0.05$. The critical values are $\pm z_{\alpha/2} = \pm 1.96$.
- Our observed test statistic is $z_{\text{obs}} = 2.1$.
- The corresponding p-value is:

$$P(|Z| > 2.1) = 2P(Z > 2.1) = 2 \times 0.0179 = 0.0358$$

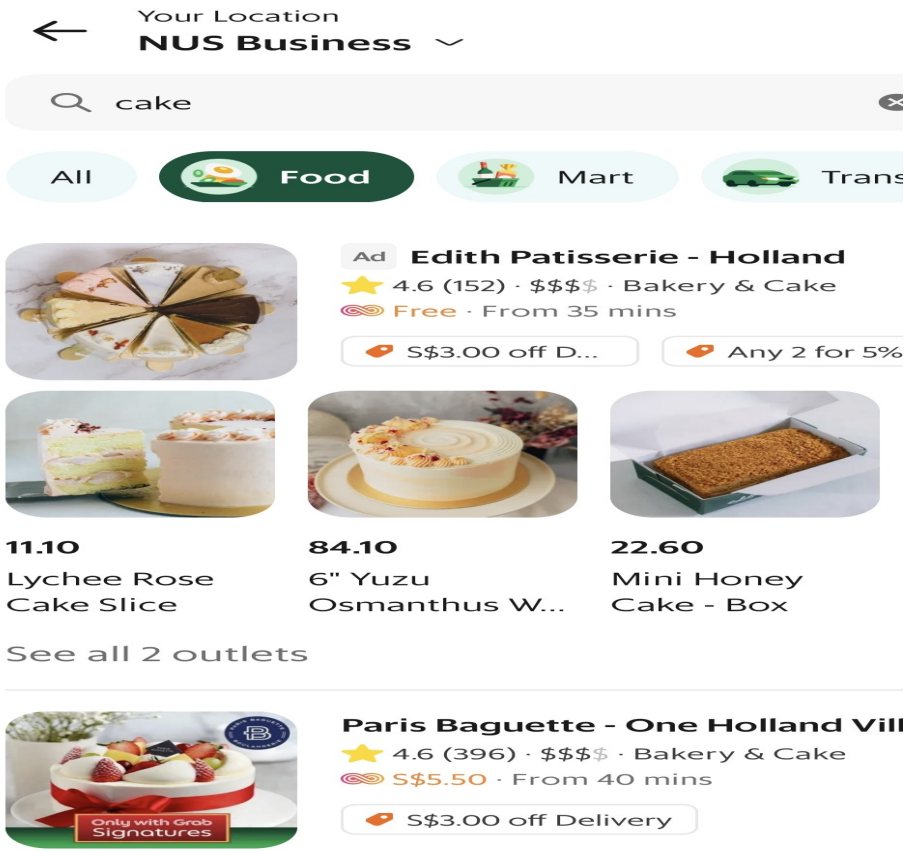
- Since $p = 0.0358 < 0.05$, we **reject** H_0 at the 5% significance level.

Alternative Distributions for A/B Testing

- We assumed that the test statistic follows a **normal distribution** (Z-test). For **large sample sizes** the CLT ensures that the Z-test is a good approximation.
- However, for **small samples** (e.g., less than 30 observations per group), the normal approximation may not be accurate.
- Instead, a **t-test** (Welch's t-test) is commonly used to account for increased variance in small samples.

Exercise

A student from NUS searches for cakes on Grab. The platform offers cakes from 4 different patisseries with the following attributes: Quality, Price, Delivery Time.



The Debate

- Data Scientist 1: Believes NUS students are highly successful and prefer **highest quality and price**.
- Data Scientist 2: Thinks NUS students are too busy and prioritize **short delivery times**.

Data-Driven Solution: Instead of debating, they decide to **run an A/B test** with two assortments.

Assortment A: Restaurants ranked by **quality** (highest first).

Assortment B: Restaurants ranked by **delivery time** (shortest first).

The Demand Model (Unknown to Grab)

Consumer Decision Process: Each consumer selects cakes based on the following utility function:

$$U_{ij} = \beta_p \cdot \text{price}_j + \beta_q \cdot \text{quality}_j + \gamma \cdot \text{delivery_time}_j + \epsilon_{ij},$$

where:

- U_{ij}: Utility of consumer i for restaurant j.
- β_p, β_q, γ: Unknown coefficients representing preferences.
- ϵ_{ij}: Random noise (Normal(0, 1)).

Assumptions:

- Consumers look at the **top 2 options** in their assigned assortment.
- Consumers purchase if $U_{ij} > 0$.
- If multiple options have $U_{ij} > 0$, they select the option with the **highest utility**.

Simulated Data for Analysis

How the Data is Generated:

- Grab does not know the true coefficients (β_p, β_q, γ).
- Data is simulated for N consumers using these values:
  - $\beta_p = -0.5$: Higher prices reduce utility.
  - $\beta_q = 1.1$: Higher quality significantly increases utility.
  - $\gamma = -0.2$: Longer delivery times slightly reduce utility.
- Restaurants Data:

Restaurant ID	Price	Quality	Delivery Time (minutes)
1	5.0	4.7	14
2	4.2	4.1	16
3	3.5	3.3	10
4	3.0	3.0	12

Purchase Decision Process

Objective:

- Use A/B test results to infer which assortment performs better.
- Metric of interest **Purchase Rate:** Fraction of consumers who made a purchase.
- In particular, how would you simulate the utilities? how would you simulate the choice probabilities? how you would implement the experiment?

Statistical Test

Purchase Rate:

$$\text{Purchase Rate} = \frac{\text{Number of Consumers Who Made a Purchase}}{n_{\text{consumers}}}$$

ATE Calculation:

$$\text{ATE} = \text{Purchase Rate}_B - \text{Purchase Rate}_A$$

Hypothesis Testing:

- $H_0 : p_A = p_B$ (no difference in purchase rates)
- Compute p-value to assess statistical significance.

2. P-Values: Interpretation, misuse, and issues in marketplaces

Points to Remember:

- A p-value does **not** measure the probability that H_0 is true.
- A p-value does **not** indicate the size or practical significance of an effect.
- P-values depend on the sample size.

Misinterpretations:

- "A p-value of 0.03 means H_0 has a x% chance of being true."
- "A smaller p-value implies a larger or more important effect size."
- "The p-value measures the probability of rejecting H_0 when it is actually true."

P-Value Misuse

- **Over-Reliance on $p < 0.05$:**
  - Results are often dichotomized as "significant" or "not significant," ignoring the magnitude or practical relevance of the effect.
  - In marketplaces, this can lead to prioritizing statistically significant but small, non-impactful changes (e.g., minor UI tweaks).

- **P-Hacking:**
  - **Stopping** data collection as soon as $p < 0.05$ is observed.
  - **Cherry-picking:** Running multiple tests and reporting only significant results.

Good Practices:

- Pre-register experiments to avoid p-hacking.
- Report effect sizes and confidence intervals alongside p-values.
- Focus on practical significance, not just statistical significance.

Continuous Monitoring A/B Testing

- In classical A/B testing, the sample size is determined **before** running the experiment.
- Example: Decide to collect data from 10,000 users and analyze results only after reaching this sample size.
- Ensures statistical tests (e.g., p-values) are valid.

Type I Error (α) – False Positive:

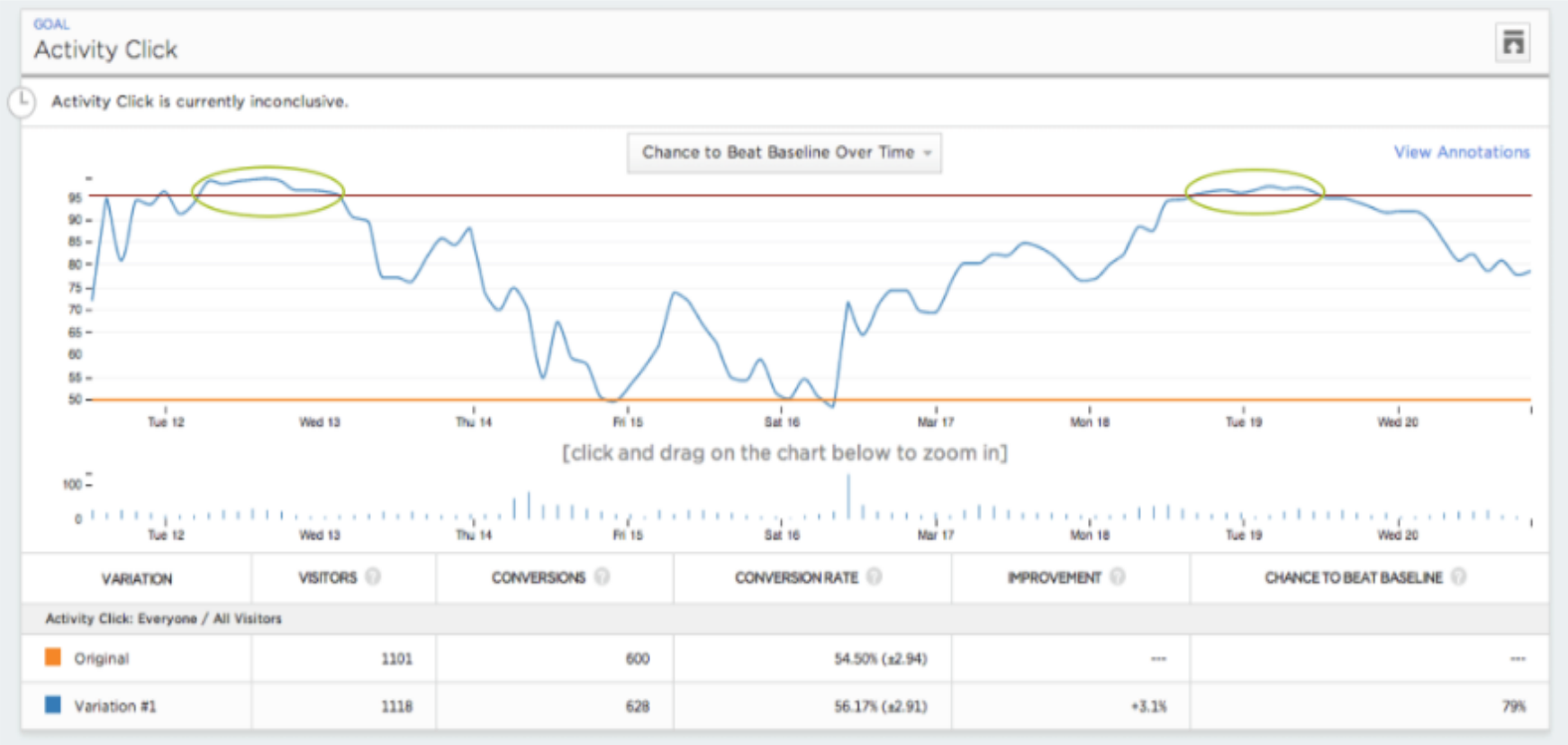
- Rejecting the null hypothesis (H_0) when it is actually true.
- Example: Declaring a new feature effective when it has no real impact.

Type II Error (β) – False Negative:

- Failing to reject the null hypothesis (H_0) when the alternative hypothesis is true.
- Example: Missing impactful feature due to insufficient data or analysis.

Issue with Continuous Monitoring

Figure shows $1 - p_n$ over time, where p_n is the p-value after n observations of the null hypothesis that the CTR in treatment and control is identical. This is A/A test: the treatment and control receive the same treatment.



The figure shows that $1 - p_n$ rises above the 95% significance threshold triggering a Type I error.

Incentives for Continuous Monitoring in Marketplaces

Repeatedly checking p-values during an experiment inflates false positive (type I error).

Why Monitor Continuously?

- **Time Pressure:**
  - Teams need quick insights to decide whether to keep or abandon new features.
  - Real-time dashboards displaying metrics like p-values.
- **Opportunity Costs:**
  - Running an experiment for too long delays other experiments.
- **Incentives:**
  - Data scientists feel pressured to declare positive results.

Mitigating Continuous Monitoring Pitfalls

Classic Fix: Commit to Predefined Timeframes:

- Choose a fixed sample size or experiment duration before starting.

Modern Fix: Use "Always Valid" Methods:

- **Always Valid P-Values:** Maintain statistical validity regardless of when the experiment stops.

e.g., Paper: "Always Valid Inference: Continuous Monitoring of A/B Tests"

Question: other solution? maybe incentives?

Multiple Testing

A/B/N Testing

- A/B/N testing is an extension of A/B testing where multiple variations $N \geq 2$ are tested simultaneously.
- Used in online marketplaces for testing multiple changes at once (e.g., UI layouts, pricing models, recommendation algorithms).

Inflated False Positive

- In A/B testing, recall that we reject the null hypothesis when $p < \alpha$.
- If we run multiple tests, the probability of at least one false positive increases. E.g., with 20 independent tests at $\alpha = 0.05$,

$$P(\text{at least one false positive}) = 1 - (1 - 0.05)^{20} \approx 0.64$$

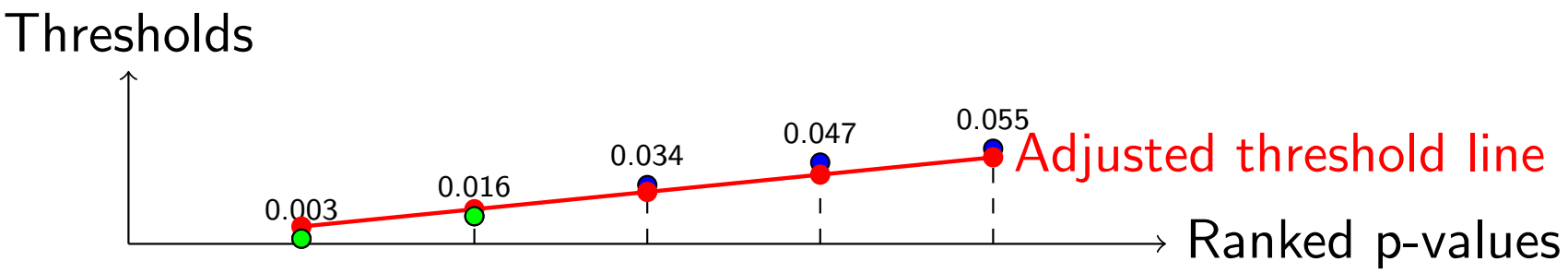
Multiple Comparisons Problem

- We need a way to control the false positive rate while maintaining statistical power.
- The False Discovery Rate (FDR) method ensures that if we declare findings as significant, a limited fraction are false discoveries.
- The most common approach: **Benjamini-Hochberg (BH) procedure:**
  1. Rank the m p-values $p_{(1)} \leq p_{(2)} \leq p_{(m)}$.
  2. Compute the threshold for each ranked p-value:

$$p_{\text{adjusted},i} = \frac{i}{m}\alpha$$

  3. Find the largest i where $p_{(i)} \leq p_{\text{adjusted},i}$.
  4. Reject all hypotheses with p-values $p_{(1)}, p_{(2)}, \ldots, p_{(i)}$.

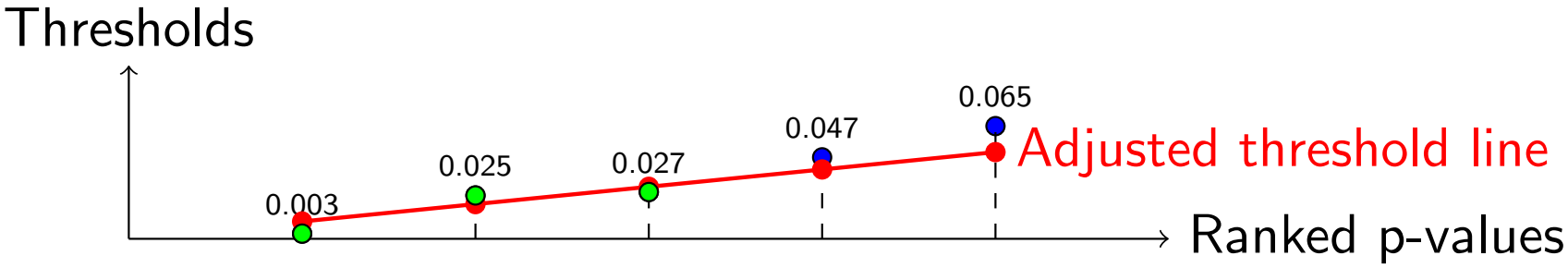
Benjamini-Hochberg Procedure



The **first two hypotheses** are **significant** (marked in green) and the rest are not rejected by BH (why)?
Without correction all p-values below 0.05 would be significant.

Rank i	p-value $p_{(i)}$	Adjusted threshold $p_{\text{adjusted},i}$
1	0.003	0.010
2	0.016	0.020
3	0.034	0.030
4	0.047	0.040
5	0.055	0.050

Benjamini-Hochberg Procedure



The **first three hypotheses** are **significant** (marked in green), despite $p_{(2)} = 0.025 > 0.020$.
The remaining hypotheses are not rejected by BH. Why?

Rank i	p-value $p_{(i)}$	Adjusted threshold $p_{\text{adjusted},i}$
1	0.003	0.010
2	0.025	0.020
3	0.027	0.030
4	0.047	0.040
5	0.065	0.050

Why Use the Adjusted Thresholds in FDR?

- The adjusted thresholds in the Benjamini-Hochberg procedure follow the formula:

$$p_{\text{adjusted},i} = \frac{i}{m}\alpha$$

  where:
  - i is the rank of the p-value (smallest is 1, second smallest is 2, etc.).
  - m is the total number of tests.
- It turns out that the formula provides an optimal balance between avoiding false positives (Type I errors) and retaining statistical power (avoiding Type II errors).

Practical Challenges of Applying BH in Online Marketplaces

- **Defining the Family of Tests?**
- **Independence Assumption?**
- **Multiple Teams?**

Correlated Tests: Amazon Example

Two Simultaneous Experiments:

- **Experiment 1: Lower Inventory Holding Costs**
  - **Metric:** Average inventory per seller.
- **Experiment 2: Lower Per Sale Fees**
  - **Metric:** Units sold per seller.

Experiments Can Be Correlated

- More inventory (Exp 1) $\rightarrow$ More competition $\rightarrow$ **Lower per-seller sales**.
- Higher sales (Exp 2) $\rightarrow$ Faster stock depletion $\rightarrow$ **Lower inventory levels**.

How to Handle This?

3. SUTVA Violations and Network Interference in Marketplaces

In online marketplaces, actions taken by one user can impact the outcomes of others, leading to SUTVA violations.

Example – Airbnb Hosts:

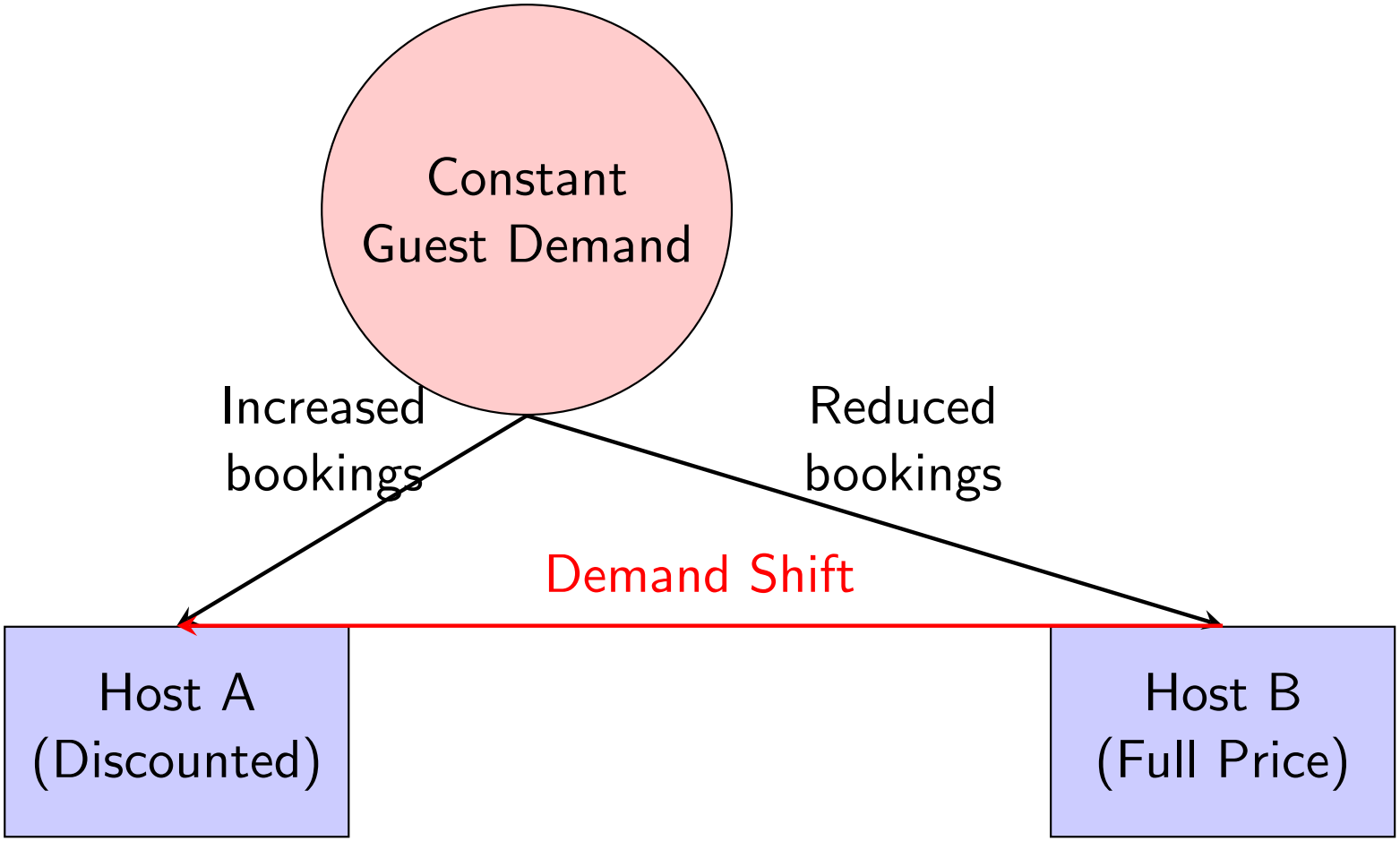
- **Scenario:** Airbnb implements a discount for certain hosts to test price sensitivity.
- **Potential Violation:** Discounted listings may reduce bookings for full-price listings due to constant total demand (*cannibalization*).
- **Implication:** The treatment assigned to one host affect other hosts, violating SUTVA.

Implications for A/B Testing:

- SUTVA violations can lead to biased estimates of treatment effects.

SUTVA Violations in Airbnb Example

- **Scenario:** Airbnb applies a 10% discount for Host A while keeping Host B's pricing unchanged.
- **Effect:** Bookings for Host A increase, but at the expense of Host B due to constant total demand (*cannibalization*).



SUTVA Violation in Airbnb Example

- **Initial Bookings Before Experiment:**
  - Host A: 10 bookings at 100 per night.
  - Host B: 10 bookings at 100 per night.
- **After Discount for Host A (10%):**
  - Host A: 15 bookings at 90 per night (5 gained from Host B).
  - Host B: 5 bookings at 100 per night.
- **Outcome:**
  - Host A bookings increased, but total demand remains the same (cannibalization).
  - Treatment effect appears positive for Host A, but it overestimates the benefit as total marketplace demand has not grown.

Takeaway

Measuring the treatment effect overestimates the intervention, as it ignores cannibalization effects.

Network Interference Examples

Network Interference and SUTVA violations are widespread in online marketplaces.

Examples? e.g., Facebook, Linkedin, Grab?

Addressing SUTVA Violations in Marketplaces

Possible Solution: Geographic Separation:

- **Approach:** Conduct experiments in distinct geographical areas to minimize interactions between treated and untreated units.
- **Example:** Test a new feature on hosts in one city (e.g., Singapore) while keeping other cities (e.g., Hong Kong) as the control group.
- **Benefit:** Reduces the risk of interference between treated and untreated units.
- **Limitations:** 1. Heterogeneity Across Locations.
- **Limitations:** 2. Loss of Statistical Power.

Airbnb Meta-Experiment Paper

Paper Title: "Reducing Interference Bias in Online Marketplace Pricing Experiments"

- Study focuses on estimating the treatment effects in online marketplaces while accounting for interference (network).
- Uses meta-experiments on Airbnb to demonstrate the impact of cluster randomization.

Meta-Fee Experiment:

- Tested changes in platform fees for long-tenured hosts in select markets.
- Experiment analyzed the causal impact of fee changes while addressing interference biases.

Cluster Randomization: Beyond Geography

Approach: Use data-driven clustering to group similar units.

- (Hopefully) minimal interaction between clusters so treatment effects are isolated.

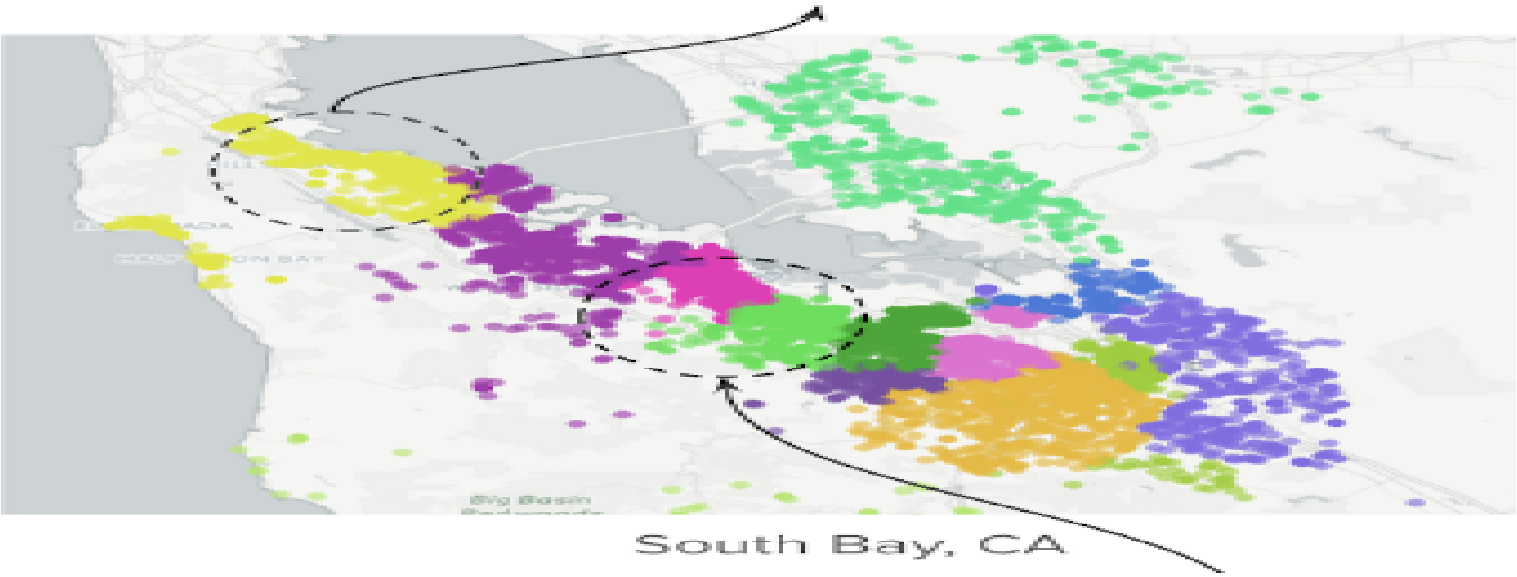
Example: Airbnb Paper

- Use embedding-based clustering methods (e.g., skip-gram models to produce 16-dimensional vector representation for each Airbnb listing based on data consisting of sequences of listings that individual users view in the same search session. Then apply recursive partitioning tree algorithm to group similar listings).
- Randomly assign entire clusters to treatment or control groups.

Meta-Fee Experiment: Design and Context

Experiment Design:

- Cluster assignments reduce interference bias by grouping listings likely to compete for the same demand.



Fee Meta-Experiment Overview

- **Dates:** March 16–21, 2019
- **Population:** 4,578,028 listings
  - **Bernoulli-Randomized :** 1,146,537 listings
    - Treatment: 573,346
    - Control: 573,191
  - **Cluster-Randomized :** 3,431,491 listings
    - 2,982 clusters were assigned to the treatment and 2,982 clusters were assigned to the control.
    - Treatment: 1,720,147
    - Control: 1,711,344
- Note: loss of statistical power

Fee Treatment Design

Treatment Mechanism:

- **Long-tenured Listings:**
  Treatment: Fees **increased**.
  Control: Fees **decreased**
- **Less-tenured Listings:** No fee changes.

Conceptual Comparison:

- Driving business to newer listings vs. rewarding long-time Airbnb hosts.

Reversal Experiment:

- Dates: April 15–22, 2019.
- Mitigate negative impacts of meta-experiment on Airbnb hosts.
- Reversed conditions: Treatment $\leftrightarrow$ Control.

Results: Treatment Effects for Meta-Fee Experiment

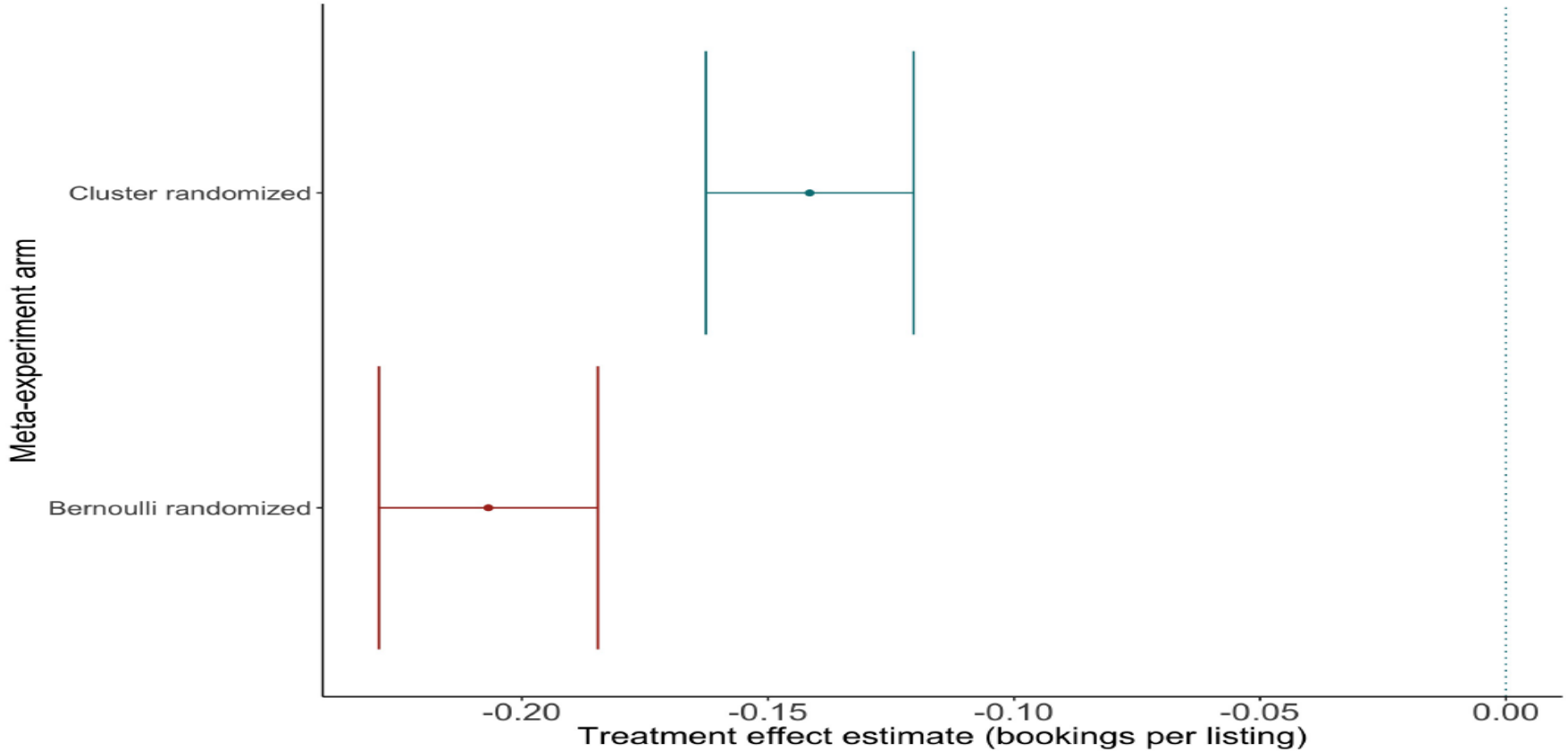


Figure 7 Total average treatment effect estimates for the fee experiment, estimated separately in the Bernoulli randomized meta-treatment arm and the cluster randomized meta treatment arm. Error bars represent 95%

- Cluster randomization provides more reliable treatment effect estimates by minimizing interference.
- Interference in Bernoulli randomization can significantly distort results.

Limitations and Takeaways

Limitations of Cluster Randomization:

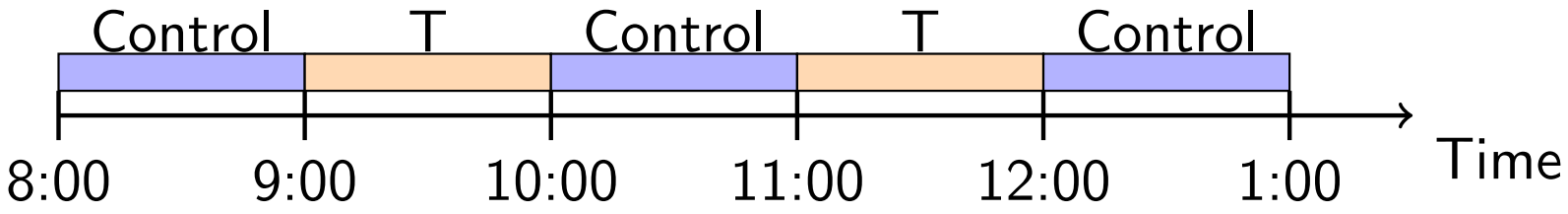
- Loss of Statistical Power:
  - Larger clusters reduce the number of independent observations.
- Clustering Method:
  - Imperfect clusters may still include listings competing for the same demand.
- Operational Complexity?

Takeaways:

- Cluster randomization outperforms standard randomization in network settings.
- Data science work is crucial for reliable results (e.g., careful design and validation of clusters, monitoring, statistical analysis).

Other Solutions: Switchback Experiments

- **Approach:** Rotate the treatment on and off over time for the same unit (e.g., all hosts in Singapore) and measure outcomes during each period.



- **Example:** Apply a visibility-boosting feature to Airbnb hosts in a city during odd weeks and remove it during even weeks.
- **Benefit:** Comparison is within the same unit.

Switchback Experiments: Limitations

- **Limitations:** 1. Carryover Effects: A visibility boost for a host may result in repeat bookings or increased reviews that continue to benefit the host even after the boost is removed.
- **Limitations:** 2. Temporal Effects: If the treatment is applied during high-demand weeks and removed during low-demand weeks, the observed effect might be due to demand variability rather than the treatment itself.

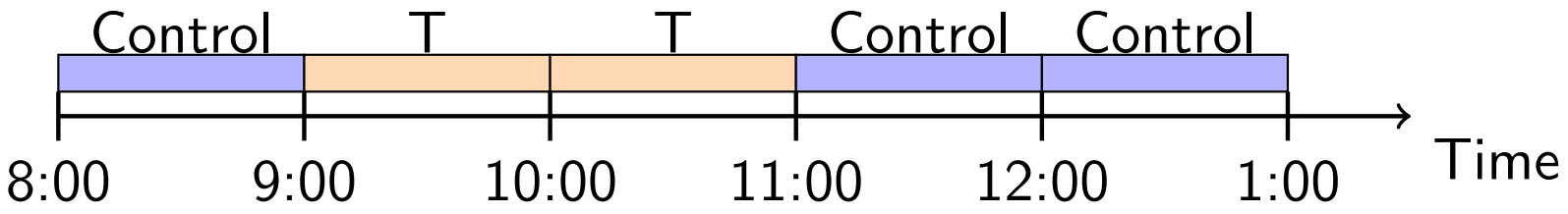
Randomization in Switchback Experiments

Approach: Randomize treatment assignments to mitigate temporal biases.

Example: Hosts in Singapore

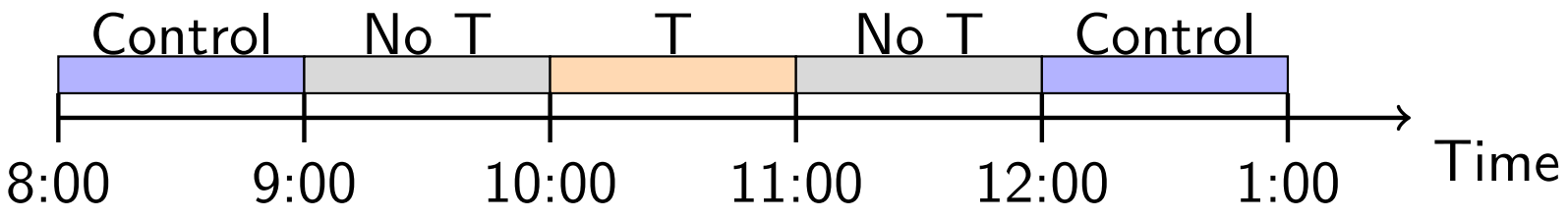
- **Week 1:** Treatment (boost visibility).
- **Week 2:** Control (no boost).
- **Week 3:** Randomly assign treatment or control again.

Ensures treatment/control conditions are independent of external trends.



Mitigating Carryover Effects

- Introduce periods with **no treatment** between control and treatment phases.
- Alternate randomly between treatment and control with sufficient "washout" periods.



Switchback Experiments: A Temporal A/B Test

Takeaway: A switchback experiment is essentially an **A/B test** where the dimension of randomization is based on **time periods** instead of individual units.

- **A/B Test:** Users are split into control (A) and treatment (B) groups.
- **Switchback:** The platform alternates between control and treatment each period of time.
- Ideal for networked marketplaces (e.g., ride-sharing, food delivery).

What may go wrong with switchback experiment in online marketplaces (except the limitations discussed)?

Strategic Users

Switchback Experiments: Pricing Example

- **Objective:** Learn the demand by alternating prices (e.g., high and low) over time.
  Possible issue?
- **Forward-Looking Behavior:**
  - At high prices: Buyers delay purchases, anticipating future discounts.
  - At low prices: Demand includes delayed purchases from earlier periods.
  - Demand at higher prices is underestimated; demand at lower prices is overestimated.
- **Resulting Bias:**
  - Observed demand fails to accurately represent the true demand.

Addressing bias?

Hands on Example

Context:

- Airbnb conducts pricing experiments.
- Two hosts (or two types of listings) for simplicity.

Setup:

- Host 1 alternates between **treatment price** ($18) and **control price** ($20).
- Host 2's price remains constant at $20.

Demand Model:

- A **logit model** estimates booking probabilities:

$$P_{\text{booking}} = \frac{1}{1 + e^{-\text{score}}}$$

- **Score:** Combination of price, reviews, and location.

Simulating Bookings

Key Parameters:

- Host 1 coefficients: $a_1 = -0.27, a_2 = 0.2, a_3 = 0.1, a_4 = 0.05$
- Host 2 coefficients: $b_1 = -0.21, b_2 = 0.15, b_3 = 0.12, b_4 = 0.08$
- **Compute probabilities:**

$$P_{\text{Host 1}} = \frac{1}{1 + e^{-(a_1 \cdot \text{price}_1 + a_2 \cdot \text{price}_2 + a_3 \cdot \text{reviews} + a_4 \cdot \text{location})}}$$

- **Simulate bookings:** Binomial random variable based on P_{booking}.
- For even days price for host 1 is 18\$ and for odd days 20\$. For host 2 price is fixed at 20\$.

Estimating the Treatment Effect

1. **Define Average Bookings:**
   - Avg. bookings (treatment) $= \frac{\text{Total bookings under treatment}}{\text{Days under treatment}}$
   - Avg. bookings (control) $= \frac{\text{Total bookings under control}}{\text{Days under control}}$
2. **Treatment Effect:**

$$\text{Avg. bookings (treatment)} - \text{Avg. bookings (control)}$$

Class Exercise

- Run the simulation to generate bookings data.
- Calculate treatment effect using the defined formulas.
- Take **8 minutes** to design your approach. In particular, what would the next few lines in the code would do?

```
import numpy as np

# Parameters for the demand model
a1, a2, a3, a4 = -0.35, 0.22, 0.2, 0.15
b1, b2, b3, b4 = -0.28, 0.22, 0.22, 0.18

# Constants
n_days = 10000
price_treatment = 18
price_control = 20
reviews = 2, 2  # x3
location = 3, 2  # x4

# Seed for reproducibility
np.random.seed(42)
```

What happens to the treatment effect when host 2 price is reduced to 18?

Results and Discussion

Simulated Results:

- Host 2 price 20: Estimated treatment effect: 0.11.
- Host 2 price 18: Estimated treatment effect 0.08.

What are the issues with this A/B test except the network issue we discussed?

4. Other Challenges: External Validity and Heterogeneity

- **External Validity:** Results from an A/B test may not generalize to other user segments, regions, or time periods.
- **Heterogeneity of Treatment Effects (HTE):** Different groups (e.g., demographics, regions) may respond very differently to the same treatment.
  Examples?

- Online marketplaces (e.g., Uber, Airbnb) serve diverse users, making uniform results less useful.

Heterogeneity in Marketplaces

Examples:

- Airbnb:
  - A discount may increase bookings in tourist-heavy cities (e.g., Paris) but have little effect in rural areas.
- Grab:
  - Surge pricing may increase supply in Singapore but has limited effect on supply and deters rides in cities like Jakarta.

- Testing one-size-fits-all treatments can mask useful subgroup insights.
- Without addressing heterogeneity, interventions may underperform in the marketplace as a whole.

Mitigating Heterogeneity in A/B Testing

Stratified or Group A/B Testing

- Divide users into groups (e.g., by region, demographics, contexts).
- Run separate experiments for each group to measure specific effects.

Challenges:

- Requires larger sample sizes for subgroup analysis.
- Increases experiment complexity and engineering resources.

What are the different goals of Group A/B Testing and Cluster Randomization?

Other Challenges in A/B Testing: Scale

- In big marketplaces there are thousands of concurrent experiments each week; many teams optimize different objectives.
- Risk: a "win" for one team can harm company-wide outcomes or another team's key metric.

Examples:

- (From Airbnb Blog Post:) If house rules are not displayed in Checkout, bookings may increase but ratings may be lower post-trip.
- Many other possible examples, e.g., fraud friction: adding extra verification could reduce fraud but also reduce bookings due to added friction.

Solution: Guardrails

- Solution: Many platforms use a company-wide Guardrails:
  - If an experiment triggers a guardrail, it is not auto-killed.
  - It enters an escalation process for transparent stakeholder review.
- Guardrail metrics (company-wide metrics such as revenue / user experience that launches should avoid substantially harming) matter for the company as a whole; a launch need not improve them, but should avoid substantial negative impact.
- Airbnb launched such a system in 2019 (Airbnb blog post).
- More guardrails $\neq$ better:
  - If you protect many metrics and alert on any degradation at $p < 0.05$, false alerts become very likely.
  - E.g., with 50 metrics, you get at least one false alert about 92% of the time in an AA test.

One-line intuition for 92%:

$$\mathbb{P}(\text{at least one false alert}) = 1 - (1 - 0.05)^{50} \approx 0.923.$$

Is the Experiment Safe to Launch?

Typically, systems have multiple guardrails an experiment must pass before it can launch.

1. Impact Guardrail:
   - Escalate (flagged for review) if the global average effect is more negative than a threshold t (e.g., revenue drops by more than t percent, or customer service tickets increase by more than t percent).
2. Power Guardrail:
   - Requires the standard error of the percent change to be small enough. Helps prevent premature decisions from early stopping / continuous monitoring.
3. Extra protection for select metrics:
   - For chosen top metric(s), escalate if the effect is negative and statistically significant (e.g., $p < 0.05$).

Adjusting for Global Coverage

- Global coverage = percent of total platform users/visitors assigned into the experiment.
- Issue: low-coverage experiments have noisier estimates and would struggle to pass a single fixed Power Guardrail.
- Marketplaces typically adjust guardrail thresholds based on an experiment's coverage so small-traffic tests aren't flagged just because they're noisier, while still protecting the business from meaningful potential harm.
- Example: A 1 percent traffic test may need a wider percent-change band to avoid noisy escalations (i.e., a wider threshold).

Escalation and Decision-Making

- Escalation $\neq$ rejection: the goal is transparent tradeoff discussion, not bureaucracy.
- Reported outcomes from Airbnb blog post:
  - Roughly 25 experiments flagged per month (2019-202).
  - About 80% eventually launch after discussion / added analysis.
  - About 20% are stopped before launch (roughly 5 per month).

Guardrails are a coordination/alignment mechanism across teams: they ensure local experimentation decisions are consistent with global marketplace health.
Other examples of alignment? Metrics.. ?

Other Challenges in A/B Testing: Algorithms

In online marketplaces, algorithms can neutralize the effects of small A/B tests, leading to no measurable impact.

Example: Autobidding in Ad Auctions

- A small-scale A/B test modifies the design of ad allocation.
- Competing autobidders, relying on large datasets and neural networks, treat the change as noise and do not adjust bids.
- Result: Experiment shows no effect.

Possible Solutions:

- **Large-Scale Tests:** Apply changes to a significant portion of the marketplace to create observable disruptions.
- **Simulation and Modeling:** Use game-theoretic or agent-based simulations to anticipate and analyze algorithmic stability.

Other issues with testing on algorithms?

Other Challenges in A/B Testing

Short Test Durations / Small Sample Size:

- Example: A one-day test may not account for weekly behavior cycles.
- Example: Testing a rare event, like high-value purchases can be problematic.
- Long term effects?

Ignoring External Factors:

- Unaccounted events can skew results.
- Example: Competitor sales during a test period.

Next Lecture: Alternative to A/B Testing

Multi-Armed Bandit (MAB) Framework:

- **Why MAB?**
  - A/B testing can be slow and risky.
  - MAB algorithms dynamically allocate treatments while learning from real-time data.
- **Contextual MABs:** Incorporate user or environment features (e.g., location, preferences) to personalize treatments.

Discussion Questions

Do you think there are diminishing returns in A/B testing?

What are the potential disadvantages of relying mainly on A/B testing for decision making in online platforms?